Adam Shebani

Operational Amplifiers Procedure

Instructional Objectives

1. Assemble a non-inverting amplifier circuit with an op-amp and measure the transfer function or gain of the output voltage with respect to the input voltage of the circuit.
2. Look at some parameters of a real op-amp.
   a) Measure the output current limit of **real** op amp.
   b) Observe clipping limitations of the op amp when used with AC or DC inputs.
3. Determine a 'reasonable' range for the resistances and gains used for op-amp circuits as dictated by the **real** op-amp.

Procedure

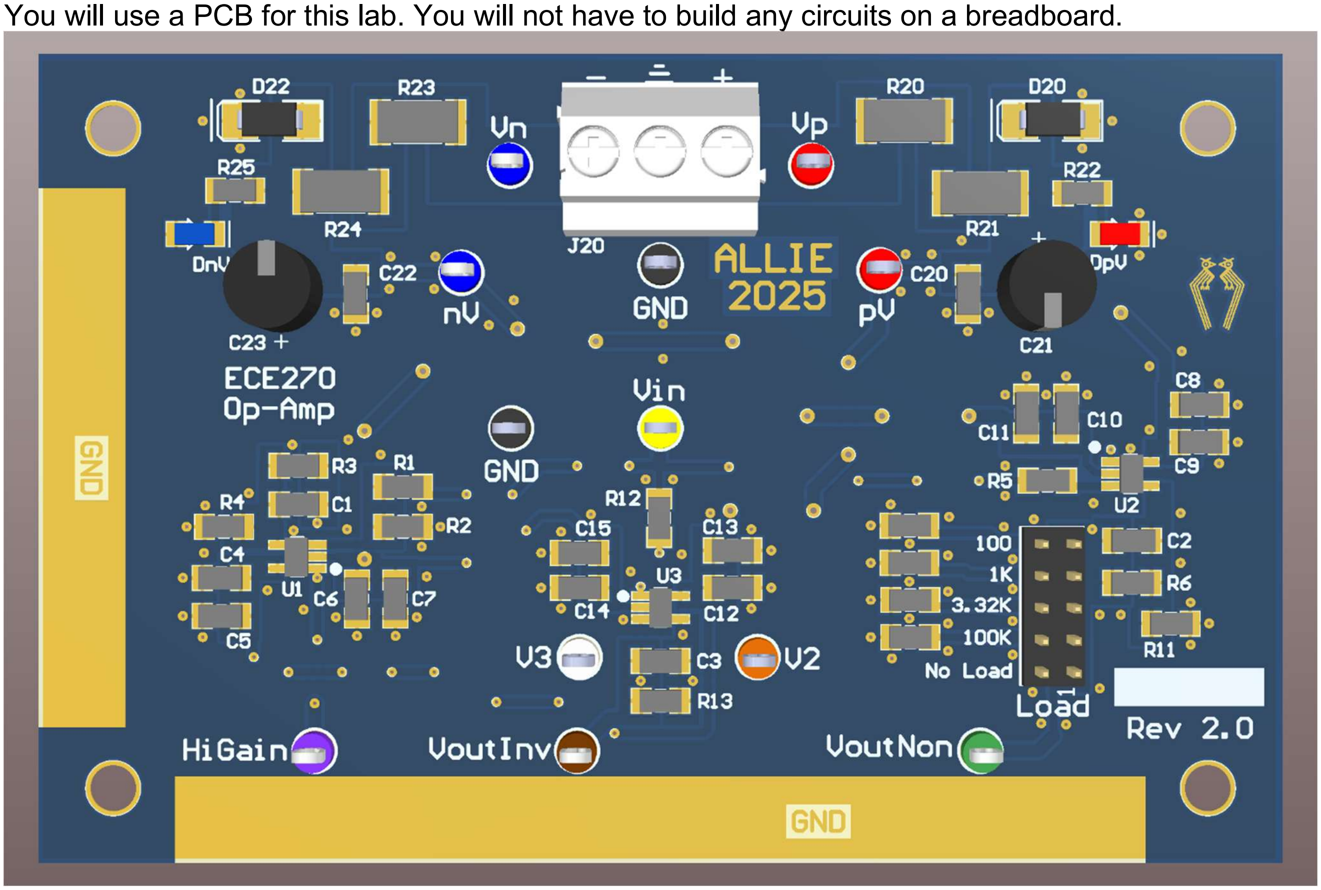
Components needed:
None: You need the ECE270 Op-Amp PCB.

0. Make sure you have looked at the Inverting and Non-Inverting amplifier configuration in the theory section.

STOP: Read the PowerSupplyDesc.docx now

It explains how to setup the power supply for this lab! It is the same connections as when you used the PCB to buffer the square wave in the RC/LC lab.

Equipment setup:
1. You will use 4 distinct pieces of equipment.
- You will use a PCB for this lab. You will not have to build any circuits on a breadboard.



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- The Power Supplies are the +30V and -30V supplies set to +5V and -5V respectively. I strongly suggest you set the current limit on both channels of the power supply to 0.025A. This will save possible damage to the board if the supplies are connected incorrectly.

The PCB won't have the white connector. Use the test points Vn, Vp and GND to input the DC power supply. The gold bands are GND. You can connect scope probe grounds to these.

- Remember that using tracking on the power supply means you only have to set the +5V voltage.
- The Signal Generator is the Wav Gen on the scope for Vin.
- Use the BNC to mini clip cable for this. No voltage yet. Red mini clip. Black clip goes to GND.
- The Oscilloscope will be used to measure up to 4 signals during lab.
  - CH1 to Vin and GND.
  - CH2 to Vout and GND.
  - CH3 and CH4 to other signals and GND.
- Enable the outputs on the supply after it is connected to the PCB with the +5, -5 V and common GND connected.
- The power supply will indicate what voltage the + and – 30V supplies are at once turned on. Make sure they are somewhere near 5V in magnitude.

Reminder: To use the PCB, you have to connect $\pm5V_{DC}$ to the V_P, V_N and GND test points. Other signals are needed as described below. See the document **PowerSupplyDesc.docx** on the canvas site to see how to setup the power supply. Did you look at **PowerSupplyDesc.docx** yet? How 'bout' now?

- Connect the RED wire, +5V, to the RED, Vp test point. Set the current limit to 0.025A.
- Connect BLUE wire, -5V, to the BLUE, Vn test point. Set the current limit to 0.025A.
- Connect the BLACK wire, GND, to the GND test point.

Ready to go now.

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The PCB has 3 distinct circuits on it. You will use one at a time.

1. The first circuit you will use is an inverting amplifier circuit as described in the theory section. Note the op-amp on the PCB has 5 pins. The schematic has the 5 pins shown with 2 distinct bodies. This is done to make the schematic easier to understand. The power supply is separated from the op-amp proper.

Inverting Amplifier with low gain. The schematic is shown in Fig. 1. Determine the theoretical gain (Theory), A_{INV} = VoutInv/Vin, of this circuit. $Gain_{INVLOW}$ = **-3.33** Note: There is a capacitor, C3, that is there to limit the bandwidth of the circuit. Don't worry about what this does. Assume it is not there when answering the questions.

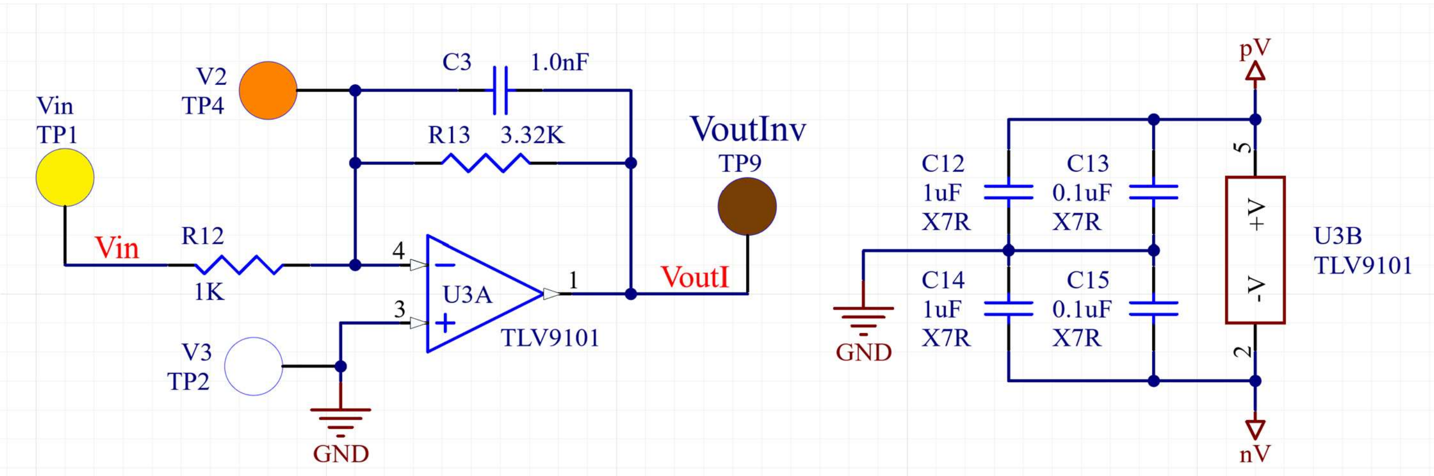

Figure 1: Inverting amplifier with low gain.

Note: The four X7R capacitors connected to pins 2 and 5 are usually referred to as bypass capacitors. The bypass capacitors are your friends. Always use them with op-amp circuits! They make the power supplies stable. One of the other benefits of using these is they prevent the op-amp circuit from oscillating (self-generating a high frequency at the output).

- Set the input voltage Vin to 2.0V_{PP} at 1KHz Sine with 0V_{DC} offset.
- What should the theoretical DC output voltage be? **0** V_{DC}.
- Note the AC input is 2.0V_{PP}.
- What should the AC output voltage, V_{OutInv}, be? **6.64** V_{PP}.
- Monitor the AC output voltage with the scope. (Peak to peak Measurement) Measure the DC output with the DMM or scope.
- Use triggering on WaveGen for the whole lab.
- Use Averaging on the Acquire menu for the whole lab. Pick between 8 and 32 averages.

2. **Measure Vin and V_{OUT}.** to make sure the circuit is working as expected. Set the time base to about 200us/div. Put the measured values in Table 1.

Review the theory section of the inverting amplifier. Make sure you understand what virtual ground means.

- Now connect the CH3 input to the V3 test point of the op-amp. V3 is connected directly to ground. The CH3 scope trace should be a straight line at 0V.
- Change the CH3 sensitivity to 1 mV/div. Is it still a straight line at 0V? **NOPE IT'S FUZZY.**
- Put the CH4 probe on test point V2 of the op-amp. Set the sensitivity to 1mV/Div. Adjust as necessary.
- Now measure V2 and V3 AC and DC voltages only with the scope.

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Table 1:

Op-Amp Gain	1KHz 2V_{PP} sine wave with 0V_{DC}							
	Vin_{AC}	Vin_{DC}	$V_{OutInvAC}$	$V_{OutInvDC}$	V_{3AC}	V_{3DC}	V_{2AC}	V_{2DC}
	1.77 vAC	**0.00345 VDC**	**5.71 VAC**	**-0.0104 VDC**	**400 mVAC**	**0 V**	**10 mVAC**	0 **VDC**

NOTE: If the output of the op-amp gets noisy try the X10 probe setting. Don't forget to change the probe setting on the scope too.

The second circuit we will use is the low gain, Non-Inverting circuit. The schematic is shown in Figure 2.

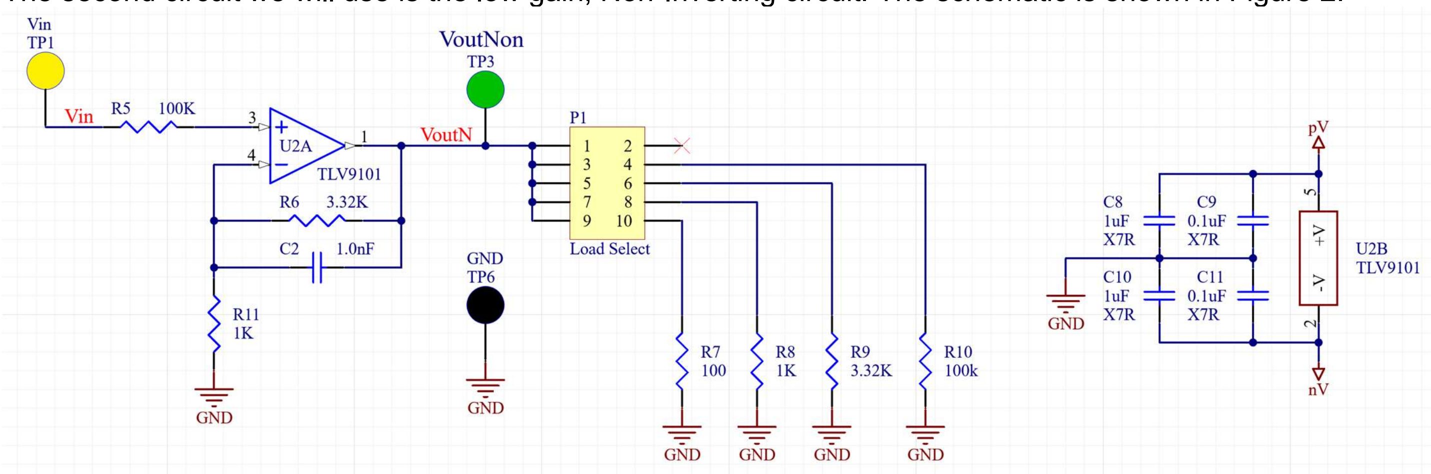

Figure 2: Schematic of the non-inverting amplifier on the PCB

- This circuit has a selectable load resistor. For now move the jumper to 'No Load' as labeled on the PCB.

Test the circuit now by putting a 1V_{PP} 1KHz Sine Wave into Vin. You should get a larger 1KHz Sine Wave out at the VoutNon test point. If not check your connections.

What is the theoretical gain, A = VoutNon / Vin, of the circuit shown in Fig.1? A_{Non} = Gain **4.32** V/V
Done checking

3. **Use the DMM or Scope** to measure Vin and V_{OutNon} for the different **DC** Vin voltages listed in Table 2 using the circuit shown in Fig. 2. Use the DMM on DC or AVG Full Screen measurement on the Scope for these measurements. Remember you must set the Wav Gen Offset to put out DC corresponding to the DC input voltages as you go. When measuring DC on the scope put the GND for the channel near the top or bottom of the screen and the DC voltage at the other end. Do this by adjusting the sensitivity (V/Div) and the display offset. Use averaging on the scope. These settings will give you the best accuracy.

V_{IN} Desired	Measured Vin	Measured V_{OutNon}	$V_{INmeasured}$*A Theoretical	V_{IN} Desired	Measured Vin	Measured V_{OutNon}	$V_{INmeasured}$*A Theoretical
0 V_{DC}	**0**	**0.0015**	**0**	2.5 V_{DC}	**2.167**	**5.485**	**9.361**
0.5 V_{DC}	**0.354**	**1.7847**	**1.530**	-0.5 V_{DC}	**-0.3533**	**-1.502**	**-1.525**
1.0 V_{DC}	**0.717**	**3.21**	**3.097**	-1.0 V_{DC}	**-0.796**	**-3.005**	**-3.439**
1.5 V_{DC}	**1.062**	**4.595**	**4.579**	-1.5 V_{DC}	**-1.196**	**-4.91**	**-5.167**

Table 2: DC voltage measurements.

What is the gain A_{Non} from above? **4.32**
Fill in the $V_{INmeasured}$*A_{Non} Theoretical column in Table 2.

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Did measured V_{OutNon} always match $V_{INmeasured}$*A? **Not Quite** What was Vin if it didn't? **Vin was large enough such that the theoretical Vout value exceeds what the op amp can produce given its sources. It got clipped.**

What you just did was measure V_{OutNon} and Vin for an op-amp circuit. 1 DC voltage at a time. This is not real efficient. There is an easier way to see Vout and Vin signals with respect to each other.

- First you should use a waveform that moves linearly from the lowest voltage of interest to the highest.
- We are in luck there is a triangle waveform that does this.
- Set the signal generator to a $3V_{PP}$ 1KHz triangle wave with 0V offset and 50% Symmetry.
- The oscilloscope probes should be connected to Vin and VoutNon.
- Set the Vin sensitivity to 0.5V/Div and the VoutNon sensitivity to 2V/Div.
- Set the time base to 200uS/Div.
- You should see the input and output voltages appear on the screen. The output should be clipped.

Please note the scope settings are a recommended start. You can change them if you want or need to.

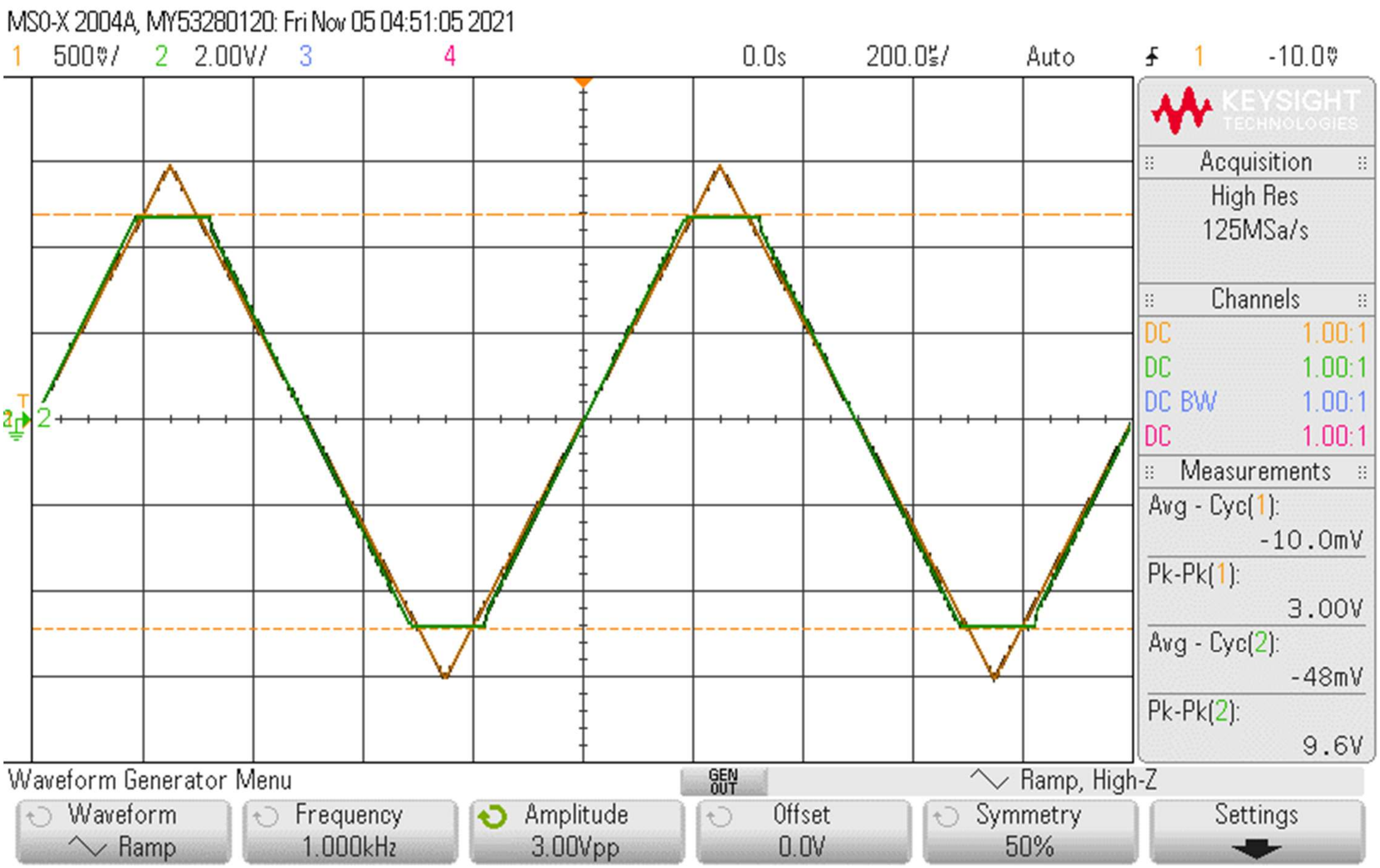

Figure 3: Triangle Vin (Yellow) and VoutNon (Green).

Next we will use the same circuit. You will be adding a load resistor. You will be changing the load resistor during the next experiment. Just add the load resistor by moving the jumper on the Load header.

4. Qualitatively observe the output resistance of the op amp circuit. This circuit is the one in Figure 2 In this case qualitative means: See if the output voltage changes when the load resistor changes. You will do this by measuring V_{OutNon} with the scope.

- Use the Oscilloscope to measure AC voltages and the DMM or scope for DC.

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- Use a DC input first and secondly use a 1V_{PP}, 1KHz **Sine** wave input for each R_{LOAD}. One at a time. Not together.
- Use R_{LOAD} as specified in table 3 for each test.
- Use the 'Wavegen' trigger on the scope to hold the sine wave still.
- Set V_{IN} to 1V_{DC} for the DC input or 2V_{PP} for a sine wave input of 1KHz. **NOTE:** Only input the AC or the DC not both simultaneously.
- The load resistors are 1% resistors. You can use the nominal value to calculate things.
- Measure the voltages at V_{IN} and V_{OutNon} as per Table 3 and enter the actual value next to the nominal value into the table. Use the Peak-peak measurement for AC and the DMM or scope AVG Full Screen for DC measurements.
- Please note the output voltages will not change much until 100Ω.

Capture V_{IN} and V_{OutNon} with the AC input for 3.3KΩ and 100Ω R_{LOAD} cases. A cell phone picture or USB key can be used for this. **See Pages 11 & 12 for these Captures!**

R_{LOAD}	DC		1KHz sine wave	
Nominal	V_{INDC}	$V_{OUTNONDC}$	V_{INPP}	$V_{OUTNONPP}$
100K Ω	1V_{DC} | **0.801**	**3.354**	2V_{PP} | **1.51**	**6.4**
3.3K Ω	1V_{DC} | **0.813**	**3.617**	2V_{PP} | **1.63**	**7.0**
1.0K Ω	1V_{DC} | **0.832**	**3.545**	2V_{PP} | **1.53**	**6.6**
100 Ω	1V_{DC} | **0.845**	**2.321**	2V_{PP} | **1.19**	**5.1**

Table 3. Measured data.

Op-amp output limit limitations: Assume the op-amp is constrained by either a voltage limit or a current limit. Both limits look like the output voltage is clipped when driving a resistor. First, we will measure the voltage limit with no load resistor. No load so the op-amp is delivering a low current. In this state we will get the highest possible output voltage for the op-amp. Then we will ask for more current than the op-amp can deliver to measure the current limit.

5. Voltage clipping limits: Set V_{IN} to a 3V_{PP}, 1KHz sine wave with no offset. You are using the Non-Inverting OpAmp circuit shown in Fig. 2, observe V_{OutNon} on the Scope. Use R_{LOAD} = ∞Ω (No Load). You will use the **scope** to measure the flattened (clipped) voltage levels.

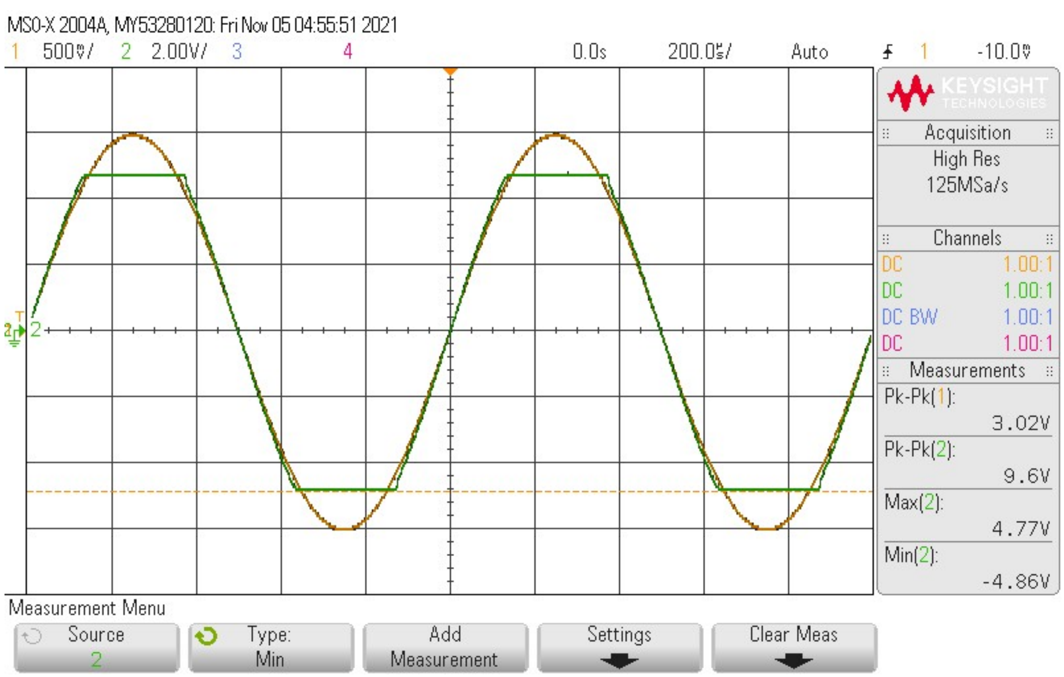

Figure 5: Clipped Voutnon display, Vin Yellow and Voutnon Green.

6. I recommend that you use the Max and Min measurement to determine V_{HIGH} and V_{LOW}. Record the positive and negative voltage levels where V_{OUTNON} flattens out. The levels will be smaller in magnitude than the power supply voltages but close. This occurs because an op-amp cannot put out a voltage outside of the power supply voltages.

   $V_{OutNonClippedHighNoLoad}$ **4.66 V** $V_{OutNonClippedLowNoLoad}$ **-4.81**

7. Output current limit: Use the circuit from Figure 2. Use an R_{LOAD} of 100Ω.

   First put a $2V_{PP}$ 1KHz sine wave in and look at the output with the scope. It should be clipping at lower absolute voltages than the voltage limits. What are the upper and lower voltage limits?

   $V_{OutNonClippedHigh100\Omega}$ **3.30 V** $V_{OutNonClippedLow100\Omega}$ **-3.46 V**

   Next set V_{IN} to $-1.0V_{DC}$. Use the **DMM** to measure V_{OUTDC}. Copy the theoretical gain calculated above down here.

   $A_{THEORETICAL}$ **4.32** V/V.

   What should the output voltage be (unclipped) for the -1.0VDC input? **-4.32 V**

   At what positive and negative voltages should the circuit clip due to the no load voltage limit measured above. ($V_{OutClippedXNoLoad}$)? pos **4.66 V**, neg **-4.81 V**.

   Measure the High voltage limit and Low voltage limit that you get when driving the 100Ω resistor. Use Vin = $+1V_{DC}$ then $-1V_{DC}$.

   $V_{IlimitOutPlus}$ **2.317 V**, $V_{IlimitOutMinus}$ **-2.375 V**.

   Are the measured High and Low voltages smaller in magnitude than the no load voltage clipping limits? Y (Y/N)

   So is the output voltage limited or current limited with the 100Ω load? I (V/I)

   Calculate the output currents by dividing the Plus and Minus voltage limits by R_{LOAD} (100Ω).

$I_{IlimitOutPlus}$ **23.1 mA**, $I_{IlimitOutMinus}$ **-23.75 mA**

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Now the 3rd circuit. The high gain circuit with HiGain as the output. This circuit has 2 parts. A resistor divider, R1 and R2, and a gain stage from op-amp U1A. Vin gets made a lot smaller by the resistor divider,Vinp. The op-amp circuit has a lot of gain to make the signal larger again.

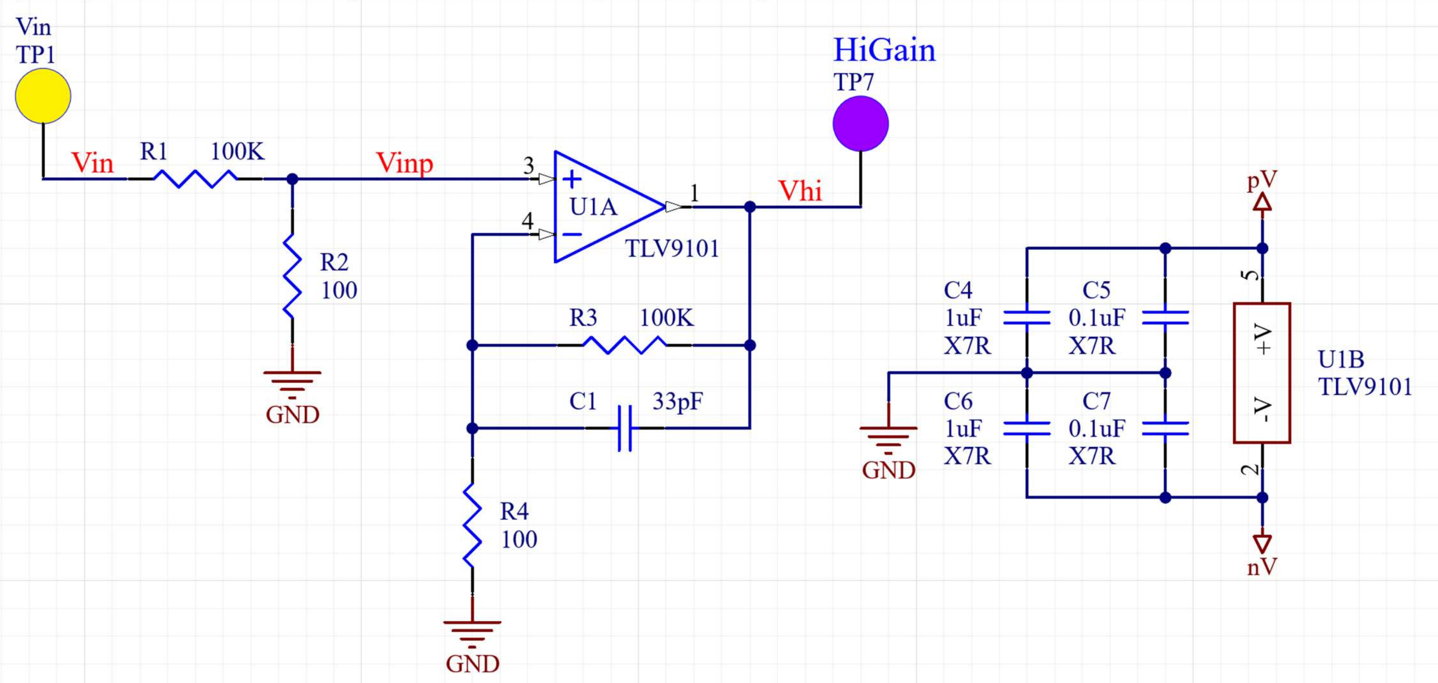

Figure 8: Non-Inverting amplifier using high gain.

What is the theoretical gain (<<1) , Vinp/Vin, of the resistor divider at the input to the op-amp circuit. $Gain_{ResDivider}$ **9.99 * 10^(-4)** Remember the input impedance of the op-amp circuit is infinite and doesn't affect the resistor divider.

What is the theoretical gain (>>1), HiGain/Vinp, of this non-inverting amplifier? $Gain_{HIGAIN}$ **1001** Don't include the resistor attenuator at the input in the gain.

For the next experiment, we want the output voltage, V_{HiGain}, to be $5V_{PP}$. What does Vinp need to be to get a $5V_{PP}$ at HiGain? **4.995 mV_{PP}**.

What does Vin (V_{PP}) need to be to get this signal at Vinp? Vin **5 V_{PP}**.

The Gain of the overall circuit in Figure 8 is $Gain_{ResDivider}$ *$Gain_{HIGAIN}$ What is $Gain_{TOTAL}$ = $Gain_{ResDivider}$ *$Gain_{HIGAIN}$ **1 V/V**

Does $Gain_{TOTAL}$ match the Vin (V_{PP}) you calculated to get HiGain (V_{PP})? **Yes, Gain_total = 1.**
If it doesn't then you need to figure out where the mistake is and re-enter your last 4 answers.

Put in a 200Hz Sine wave with the appropriate amplitude at Vin to get $5V_{PP}$ out at HiGain?

Measure Vin and HiGain with the scope. Adjust the sensitivity to give large traces.

My Vin and HiGain looked like the following.

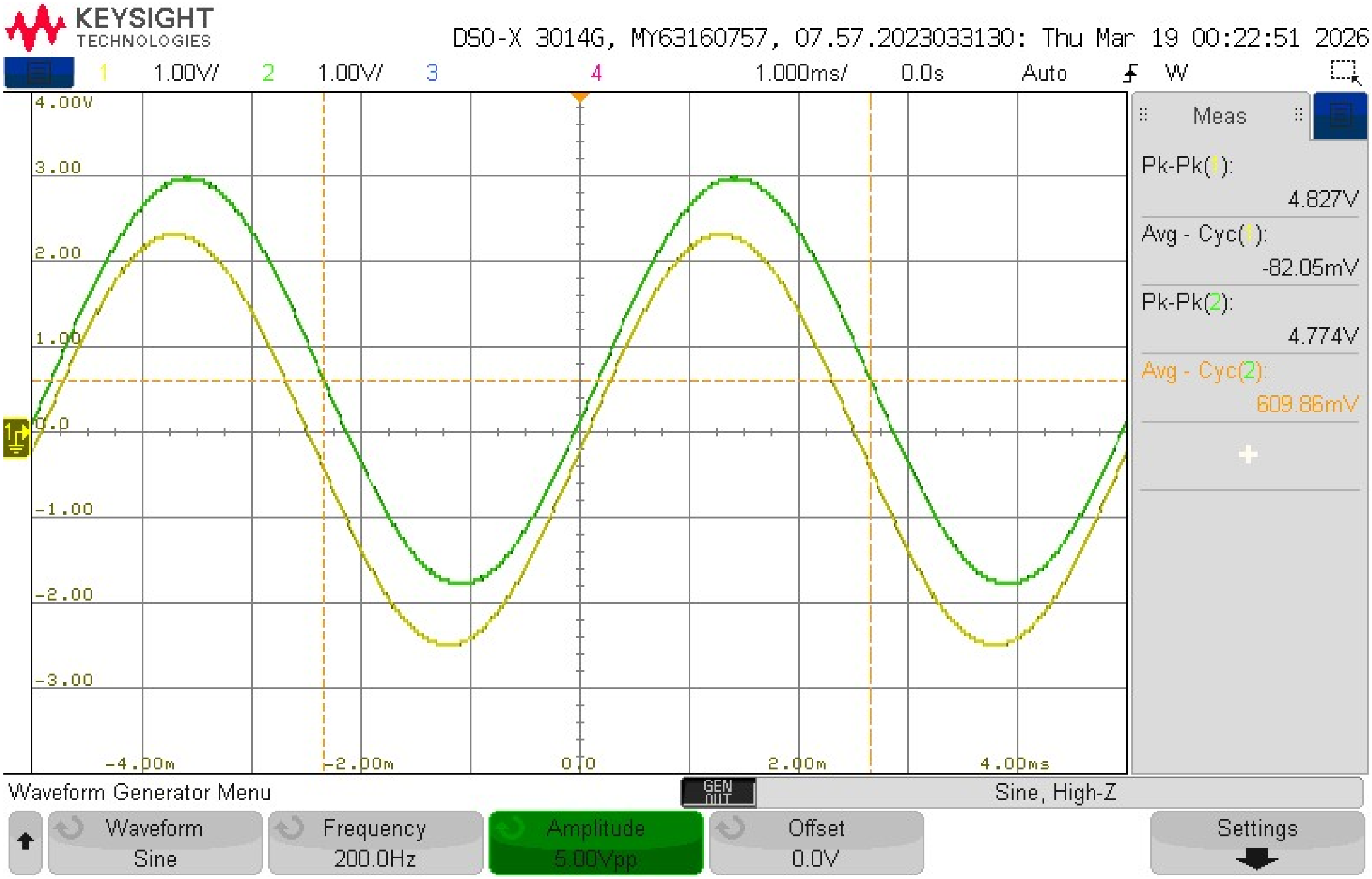

Figure 9: High Gain amplifier with attenuated input. Vin and HiGain.

Is there anything suspicious about the waveform of HiGain for the HIGH gain amplifier? Does it look different than it should in theory (Consider both the AC and DC output)? **Yes**
If yes what is “funny”?

The HiGain waveform appears to have a DC offset and is not centered exactly where ideal theory would predict. A small phase shift may also be present, but the main suspicious feature is the DC offset.

There is probably a DC offset on the HiGain signal. Let’s see where that DC output voltage is coming from?

Measure the DC of VHiGain using the DMM. $HiGain_{DC}$ **-0.062V**
Measure the DC of Vin using the DMM. Vin_{DC} **-0.071V**

Measure the DC of VHiGain using the Scope (Average Cycle). $HiGain_{DCScope}$ **-0.057V**
Measure the DC of Vin using the Scope (Average Cycle). $Vin_{DCScope}$ **-0.077 V**

What is the $Gain_{TOTAL}$ of the circuit? **1.00**

Use the DMM measurement. What is Vin_{DC} * $Gain_{TOTAL}$? **-0.071 V**

Use the DMM measurement. Is $HiGain_{DC}$ close to this value? **SORTA**

Use the DMM measurement. What is $Vin_{DCScope}$ * $Gain_{TOTAL}$? **-0.077 V**

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Use the DMM measurement. Is $HiGain_{DCScope}$ close to this value? **NOT REALLY**

They are usually not.

You have discovered one of the problems with using op-amp circuits with very high gain. In a real op-amp there is a small amount of dc offset between the In+ and In- terminals. With low gain circuits it doesn't really cause trouble. With high gain circuits it often causes trouble. To see what the DCoffset at the input terminals is you just take the $HiGain_{DC}$ from table above and divide it by the gain. So if you got 1V DCoffset with a gain of -1000 the calculated input referred offset is 1V/(-1000) or -1 mV. We call this the input referred offset. What is the measured input referred offset for your -1000 gain amplifier? **-62 microVolts**
What does the data sheet say the offset can be? You will find this in Electrical Characteristics as Vos, Input offset Voltage as a typical and a maximum voltage. Vos Typical **300 microVolts** Max **1.5 mV**

Does your op-amp meet this spec. Unless you buy a special op-amp (expensive) that has very low input referred offset like say 10uV you will have a large DC offset in an op-amp when high gain is used.
Yes it does meet the spec!

Post Lab Questions

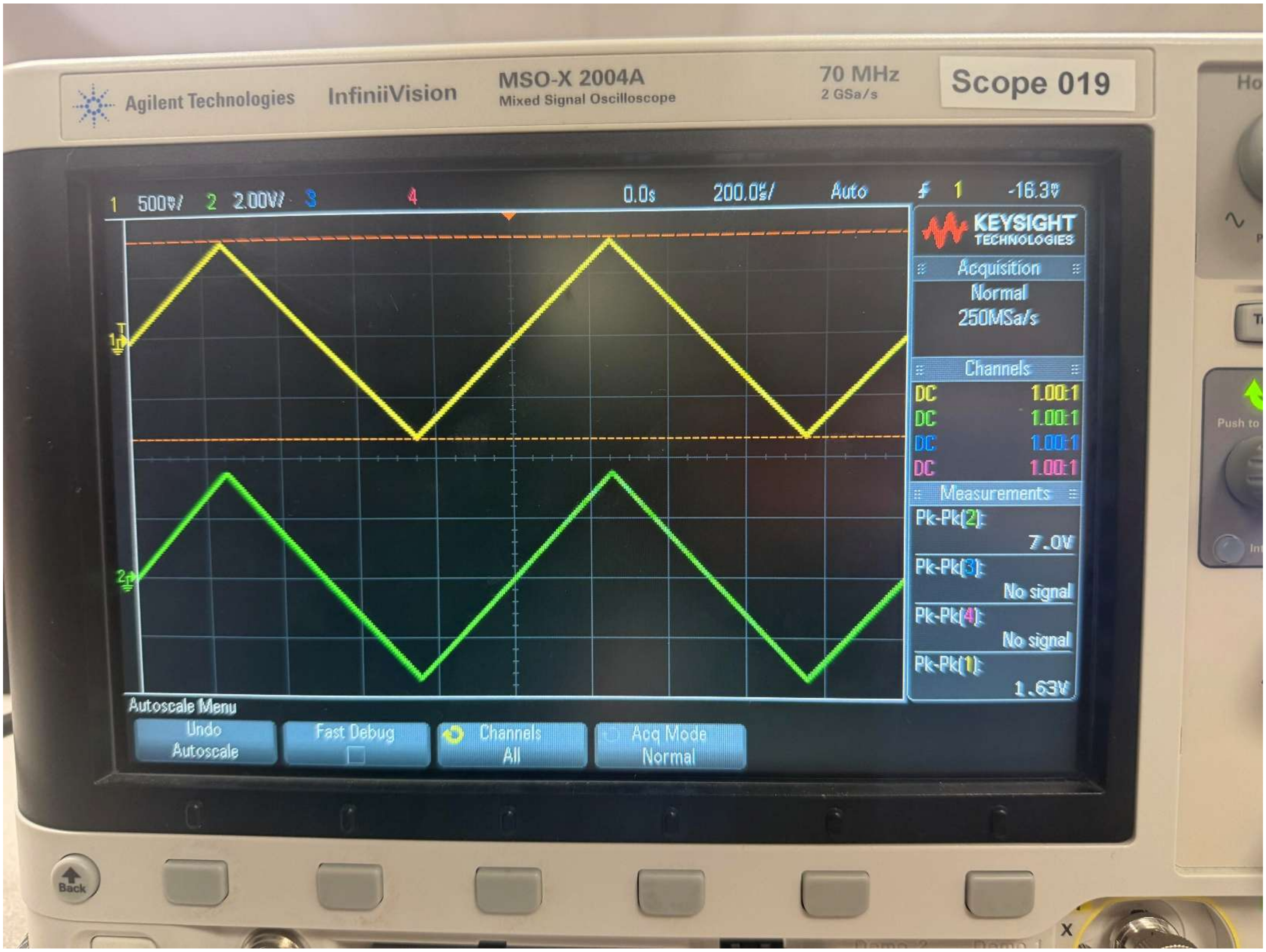
1. Plot the transfer function of V_{OUT} versus V_{IN} for the values you obtained in Table 2. Explain what is happening in the different regions of your plot.
   The V_out vs V_in plot has three regions: negative saturation, a linear region with slope ~ 4.32, and positive saturation. In the middle region, V_out ~ 4.32* V_in. At large positive and negative inputs, the output clips at the supply-limited voltage levels and the curve flattens.

2. What are all the paths that the current coming out of or going into the output (pin 1) of the op-amp for the circuit shown in Fig.2?

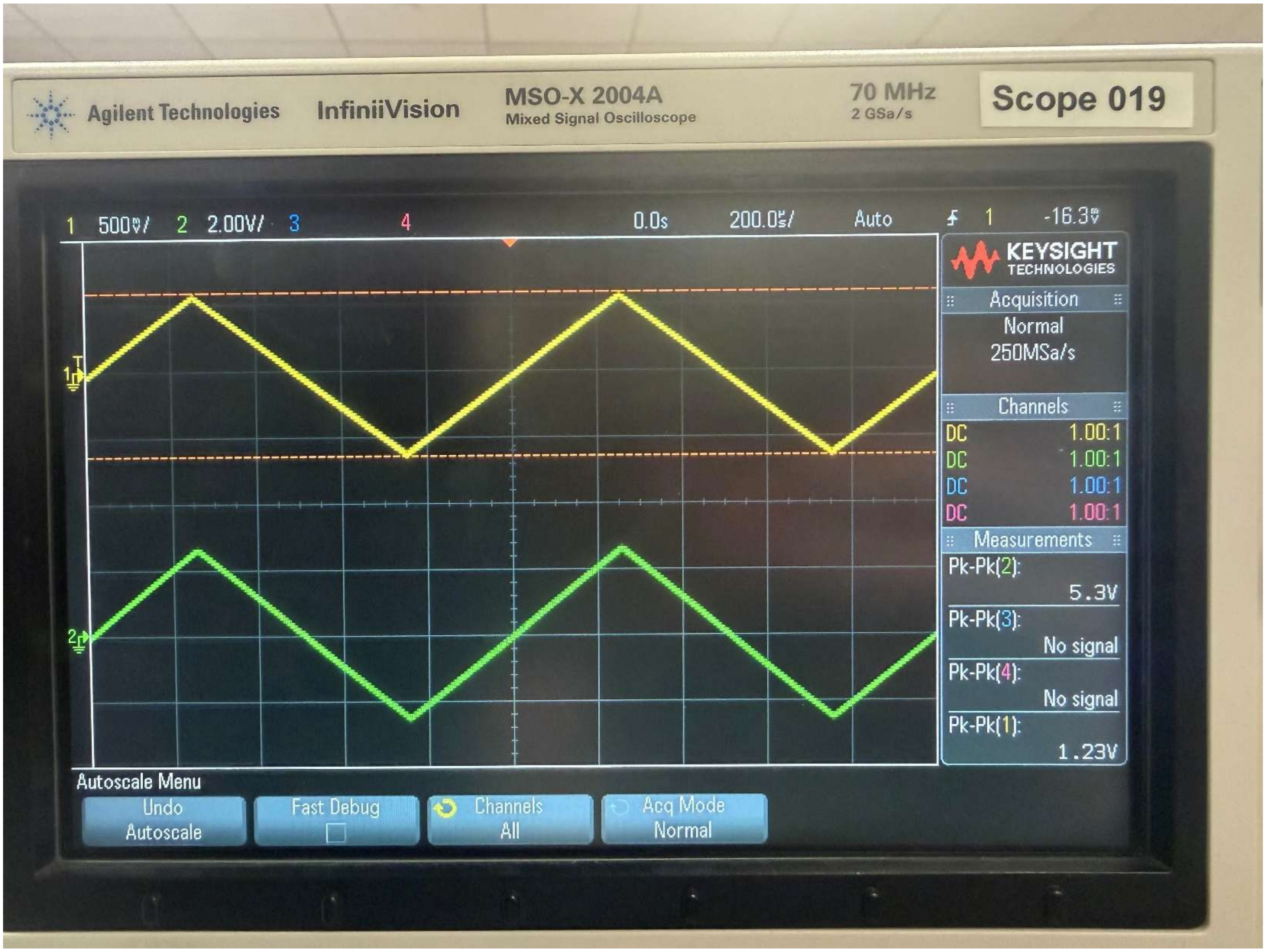
   Through the feedback resistor R_6, through the selected load resistor to ground, and in a very small amount into the measuring instruments.

3. What is the minimum load resistance (smallest resistive load) you can connect to the output so the current limit is reached just as the voltage clipping limit is reached? Choose the larger resitance of the positive or negative limits for the answer. **R_pos = 4.66/0.0231 = 201.7 ohms : R_neg = 4.81/0.02375 = 202.5 ohms. So, about 203 ohms.**

4. Is Vin_{DC} about the same as $Vin_{DCScope}$? **Yes** Is $HiGain_{DC}$ about the same as $HiGain_{DCScope}$? **Yes**
   Which of these voltages should be more accurate, DMM or Scope? **DMM**



3300 Ohm Load


Agilent Technologies
InfiniiVision
MSO-X 2004A
Mixed Signal Oscilloscope
70 MHz
2 GSa/s
Scope 019
2.00V/
0.0s
Auto
-16.3
KEYSIGHT
TECHNOLOGIES
Acquisition
Normal
250MSa/s
Channels
DC 1.00:1
DC 1.00:1
DC 1.00:1
DC 1.00:1
Measurements
Pk-Pk(2):
5.3V
Pk-Pk(3):
No signal
Pk-Pk(4):
No signal
Pk-Pk(1):
1.23V
Autoscale Menu
Undo Autoscale
Fast Debug
Channels All
Acq Mode Normal

100 Ohm Load